SIYA KOPPIKAR

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EDUCATION

Master of Science in Data Science | New York University | New York City, NY — Sep 2024 – May 2026
- Coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Big Data, Probability & Statistics

Bachelor of Technology in Computer Science & Engineering | MIT-WPU | Pune, India — Sep 2020 – May 2024

PROFESSIONAL EXPERIENCE

Software Developer Intern | FairShare IT Services Pvt. Ltd. | Pune, India — Jan 2024 – Jun 2024
- Built web apps in Python, C#, Laravel, and PHP, including a document-processing site that automates upload, parsing, and metadata removal.
- Supported the full development lifecycle from requirements and architecture through integration, testing, and QA.

Business Analyst Intern | Erisha E-Mobility | Remote, India — Aug 2022 – Nov 2022
- Analyzed market and consumer data in Python, Excel, and Power BI to support risk and supply-chain planning.
- Built recurring dashboards to surface operational bottlenecks and inform stakeholder decisions.

TECHNICAL PROJECTS

Semantic Entropy for LLM Hallucination Detection | *PyTorch, Hugging Face, Qwen2.5, DeBERTa* — Oct 2026 – Present
- Reproduced semantic entropy (Farquhar et al., Nature 2024) from scratch, clustering 10 sampled answers per question by NLI entailment.
- Detected hallucinated answers with 0.80 AUROC on TriviaQA (Qwen2.5-3B), with 95% bootstrap CIs.
- Showed semantic entropy beats naive entropy by +0.076 AUROC on sentence answers but not short ones, isolating paraphrasing as the driver.

Market-Aligned LLM Trading Signals | *LLaMA 3.2, PyTorch, Hugging Face, LoRA/PEFT* — Sep 2025 – Dec 2025
- Reframed news-based stock signals as RL from market feedback, rewarding LLM signals by realized 3-day returns, not sentiment labels.
- Built a custom Gymnasium environment and LoRA + REINFORCE fine-tuning pipeline for LLaMA 3.2-3B with 4-bit quantization.
- On a temporal holdout, an RLMF-tuned LLaMA beat the base model: 58.2% vs 53.5% win rate, 408.8% vs 248.2% cumulative return.

Real-Time ICU Anomaly Detection | *Numba, Cython, CUDA, FastAPI, PostgreSQL* — Feb 2026 – Apr 2026
- Built a real-time ICU vital-sign anomaly pipeline (Z-score/IQR + Isolation Forest) with FastAPI and live Supabase alerts.
- Benchmarked nine kernels across NumPy, Numba, Cython, and CUDA, cutting latency 857× vs. pure Python at 50K patients (4.2 s to < 5 ms).

Self-Supervised Computer Vision with MoCo v2 | *PyTorch, ResNet-50, MoCo v2* — Sep 2025 – Dec 2025
- Pretrained a ResNet-50 MoCo v2 encoder (momentum encoder, 65K negative queue) on ~500K unlabeled images; 69.7% top-1 on MiniImageNet test.
- Linear-probed frozen features on CUB-200, MiniImageNet, and SUN397; showed 128 px inference upsampling improves all three.

Credit Risk Modeling for Loan Underwriting | *Python, Scikit-learn, pandas, NumPy, SciPy* — Oct 2025 – Dec 2025
- Built a 1-year default model on 1M firm-years with a leakage-safe target and quantile-transformed ratios; 3rd in class.
- Reached ~0.80 AUC across 2008–2011 walk-forward years and 0.917 on the 2012 holdout; built the scoring harness.

SKILLS

- **Languages:** Python, SQL, Java, JavaScript, R, PHP
- **ML & LLMs:** PyTorch, Hugging Face, LoRA/PEFT, Scikit-learn, OpenCV, RL, Contrastive Learning, LLM Evaluation
- **ML Engineering:** Feature Engineering, Model Validation, Benchmarking, FastAPI, CUDA, Numba, Cython
- **Data & Tools:** PySpark, Spark SQL, HDFS, Parquet, ETL Pipelines, Git, Linux, GCP, Jupyter, VS Code